VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS304

Design and Analysis of Algorithms - Laboratory Manual & Practical Guide

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Syllabus Focus: CS304 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Algorithmic Frameworks & Divide-and-Conquer

" Syllabus Scope: Recurrence solving Master Theorem, recursion tree, MergeSort, QuickSort, Strassen matri

Unit 2: Greedy Method & Optimal Substuctures

" Syllabus Scope: Fractional knapsack, Huffman encoding tree, job sequencing with deadlines, optimal merg

CS304: Design and Analysis of Algorithms - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Dynamic Programming

" 0/1 Knapsack, Longest Common Subsequence LCS, Matrix Chain Multiplication, Floyd-Warshall all-pairs shortest

Unit 4: Backtracking & Branch-and-Bound

" N-Queens problem, Subset Sum problem, Graph Coloring, Hamiltonian Cycle, 15-puzzle state space search

Unit 5: Computational Complexity & NP-Theory

" P, NP, NP-Complete, NP-Hard classes, Cook-Levin theorem, reductions 3-SAT to Vertex Cover, Clique, Subset Su

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Comparative Empirical Timing Analysis of Divide-and-Conquer Sorting Algorithms

Lab 2: Huffman Optimal Prefix Code Generator and Text Compressor Implementation

Lab 3: Dynamic Programming: 0/1 Knapsack and Matrix Chain Multiplication

Lab 4: N-Queens Problem Solver using Backtracking with State-Space Pruning

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Thomas H. Cormen, Charles E. Leiserson - Introduction to Algorithms MIT Press 3rd Ed

* Ellis Horowitz, Sartaj Sahni - Fundamentals of Computer Algorithms Galgotia

Authorized NoteVault Educational Document " Verified for Registered University Students